

“Data Analysis” Exam
MLDM and iPSRS Master programs

December 2024 - (2 hours)

Pocket calculators and two 2-sided papers of handwritten notes are allowed.

Exercise 1 (4 points): Probabilities

The grade X on a data analysis exam is assumed to follow an unknown distribution $f_{\mu,\sigma}$ with a theoretical mean $\mu = 12$ and standard deviation $\sigma = 3$. Let us consider a set $x_1, \dots, x_{36}$ of $n = 36$ observed grades i.i.d. according to $f_{\mu,\sigma}$.

1. Let $\bar{X}$ be the empirical mean computed from the 36 observations. Compute the probability $P(\bar{X} > 13)$. **Justify your approach.**
2. How many grades n would have been needed for this probability to be equal to 1% with the same mean μ and standard deviation σ ?

Exercise 2 (6 points): Likelihood Maximization

Let $x_1, x_2, \dots, x_n$ be the observations of n random variables $X_1, X_2, \dots, X_n$ i.i.d. according to the following distribution $f(x_i; \theta)$ which depends on an unknown theoretical parameter $\theta > 1$:

$$f(x_i; \theta) = \theta x_i^{-(\theta+1)}, \forall x_i \geq 1,$$

1. Show that $f(x_i; \theta)$ is an actual density function on the domain $[1; +\infty]$.
2. Compute the expected value $E(X)$.
3. Using the likelihood maximization, find the estimate $\hat{\theta}$ of the theoretical parameter θ . For the sake of calculation, you might make use of the log-likelihood.

Exercise 3 (8 points): Regression and PCA

The first goal here is to predict the time (in hours) it takes for athletes to complete a marathon based on the number of hours they train weekly. We collect the following data from three athletes:

Weekly Training Hours (X)	Marathon Time (Y) (hours)
4	5
8	4
12	3

1. Solve the linear regression problem $Y = \theta_1 X + \theta_0$ using the close form solution.
2. Compute the mean squared error of the resulting learned regressor from the 3 athletes.
3. **(This question is independent from the first two)** Considering now that X and Y are two features, perform a PCA by:
 - (a) Computing the unit eigenvector u corresponding to the largest eigenvalue λ_{max} .
 - (b) Drawing u and projecting orthogonally the 3 original points onto this first component.
 - (c) What do you conclude?

Exercise 4 (2 points): Multiple Choice Questions

Circle the letter corresponding to the correct answer (**only one is correct**).

- Each correct answer **adds 1/2**.

- Each incorrect answer **subtracts 1/4**.

1. Let $A = \begin{pmatrix} 1 & 3 \\ 2 & 0 \end{pmatrix}$ and $B = \begin{pmatrix} 3 & 3 \\ 0 & 2 \end{pmatrix}$ be two square matrices. $(AB)^T$ is equal to:

a. $(AB)^T = \begin{pmatrix} 3 & 9 \\ 6 & 6 \end{pmatrix}$

b. $(AB)^T = \begin{pmatrix} 3 & 6 \\ 9 & 6 \end{pmatrix}$

c. $(AB)^T = \begin{pmatrix} 3 & 3 \\ 6 & 9 \end{pmatrix}$

2. The inverse of the matrix $A = \begin{pmatrix} 1 & 2 \\ 3 & 2 \end{pmatrix}$ is:

a. $A^{-1} = \begin{pmatrix} -1/2 & 1/2 \\ -3/4 & -1/4 \end{pmatrix}$

b. $A^{-1} = \begin{pmatrix} 1/2 & -1/2 \\ 3/4 & -1/4 \end{pmatrix}$

c. $A^{-1} = \begin{pmatrix} -1/2 & 1/2 \\ 3/4 & -1/4 \end{pmatrix}$

3. What is the expected value of a random variable X distributed according to the following density:

$$f(x) = \frac{1}{\theta}, \text{ for } x \in [0, \theta]$$

a. $E(X) = \theta$

b. $E(X) = \frac{\theta}{2}$

c. $E(X) = \frac{\theta}{3}$

4. What is the probability to obtained $X = 3$ “heads” when we flip a fair coin 4 times?

a. $P(X = 3) = 1/2$

b. $P(X = 3) = 1/4$

c. $P(X = 3) = 1/8$